

CCNA 200-301 Notes

Chapter 34 — Inter-VLAN Routing (Router-on-a-Stick)

Author: Yousef Sorour

This PDF follows the supplied video summary and the topology screenshot. Example addressing is based on the visible labels; verify interface numbers and VLAN 5 addressing against your actual Packet Tracer file.

1. Core Concept

VLANs separate devices into different Layer 2 broadcast domains. Devices in different VLANs/subnets need Layer 3 routing to communicate. Router-on-a-Stick uses one physical router interface with multiple sub-interfaces—one per VLAN.

PC-facing ports are Access ports. The switch-to-router uplink is a Trunk. Each router sub-interface uses 802.1Q encapsulation and an IP address that serves as that VLAN's default gateway.

2. Topology Addressing

VLAN	Department	Subnet / example gateway
2	HR	192.168.1.0/24 — 192.168.1.1
3	Sales	192.168.2.0/24 — 192.168.2.1
4	Accounting	192.168.3.0/24 — 192.168.3.1
5	Servers	Subnet not visible in the screenshot; use a separate subnet if configured.

3. Switch Configuration

Create VLANs:

```
enable
configure terminal
vlan 2
name HR
vlan 3
name Sales
vlan 4
name Accounting
vlan 5
name Servers
```

Assign PC-facing ports to their correct access VLAN. These ranges are examples only; use the actual port mapping from your topology:

```
interface range fa0/1 - 3
switchport mode access
switchport access vlan 2
```

```
interface range fa0/4 - 6
switchport mode access
switchport access vlan 3
```

```
interface range fa0/7 - 9
switchport mode access
switchport access vlan 4
```

Configure the switch port connected to the router as a trunk (the screenshot appears to show Fa0/24):

```
interface fa0/24
switchport mode trunk
end
show interfaces trunk
```

```
show vlan brief
```

4. Router-on-a-Stick Commands

The physical interface must be enabled. Create one sub-interface per VLAN. The suffix (.1, .2, .3) is a locally chosen sub-interface number; the VLAN ID in encapsulation dot1q must match the actual VLAN.

```
enable
configure terminal
interface g0/0/0
no shutdown

interface g0/0/0.1
encapsulation dot1q 2
ip address 192.168.1.1 255.255.255.0

interface g0/0/0.2
encapsulation dot1q 3
ip address 192.168.2.1 255.255.255.0

interface g0/0/0.3
encapsulation dot1q 4
ip address 192.168.3.1 255.255.255.0

end
show ip interface brief
```

VLAN 5 needs its own sub-interface and a unique subnet if you want it routed. Its subnet was not visible in the screenshot, so it is not guessed here.

5. Command Purposes

Command	Purpose
vlan 2 / name HR	Create/name a VLAN.
switchport mode access	Set a PC-facing port to Access mode.
switchport access vlan 2	Assign the port to VLAN 2.
switchport mode trunk	Allow multiple VLANs across the switch uplink.
interface g0/0/0.1	Enter/create a logical router sub-interface.
encapsulation dot1q 2	Associate the sub-interface with tagged VLAN 2 traffic.
ip address ...	Set the Layer 3 gateway address for that VLAN.
no shutdown	Administratively enable the physical interface.
show vlan brief	Verify VLANs and access-port membership.
show interfaces trunk	Verify trunk state and VLANs carried.
show ip interface brief	Verify interface IP addresses and status.

6. Optional DHCP Configuration

The video describes using the router as a DHCP server. Configure a separate pool per VLAN, using the correct network and default gateway. This is an example for the three visible VLANs; adjust excluded addresses to suit your lab.

```
ip dhcp excluded-address 192.168.1.1 192.168.1.20
ip dhcp excluded-address 192.168.2.1 192.168.2.20
ip dhcp excluded-address 192.168.3.1 192.168.3.20
```

```
ip dhcp pool HR
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
```

```
ip dhcp pool Sales
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
```

```
ip dhcp pool Accounting
network 192.168.3.0 255.255.255.0
default-router 192.168.3.1
```

7. Example End-Device Addressing

VLAN	Example PC IP	Mask	Default Gateway
HR / 2	192.168.1.10	255.255.255.0	192.168.1.1
Sales / 3	192.168.2.10	255.255.255.0	192.168.2.1
Accounting / 4	192.168.3.10	255.255.255.0	192.168.3.1

These are examples, not a claim about your current device settings. Each VLAN must use a unique subnet, and every PC's default gateway must be the router sub-interface IP for its own VLAN.

8. Verification & Troubleshooting

- On the switch, run show vlan brief and verify PC port membership.
- On the switch, run show interfaces trunk and verify the router uplink is trunking and carries the required VLANs.
- On the router, run show ip interface brief and check physical/sub-interface status.
- Ping each PC's own default gateway first, then ping between different VLANs.
- If same-VLAN ping works but inter-VLAN ping fails, check dot1q VLAN IDs, trunk status, gateway addresses, and unique subnets.
- Ensure the switch-to-router port and router physical interface are connected to the intended interfaces.

9. Exam Notes

- Router-on-a-Stick uses one physical router interface and multiple sub-interfaces.
- The switch-to-router link must be a trunk.
- encapsulation dot1q must match the VLAN ID.
- Each VLAN needs its own subnet for routed communication.
- The PC default gateway is the router sub-interface IP for its VLAN.
- DHCP is optional; use a separate pool for each VLAN subnet.

10. Quick Revision

PC → Access VLAN → Switch trunk → Router physical interface → sub-interface + dot1q VLAN ID + gateway IP → routing between subnets.

Scope note: The topic list follows the supplied video summary. HR, Sales, and Accounting subnets are based on visible topology labels. VLAN 5's subnet was not visible and has not been invented.